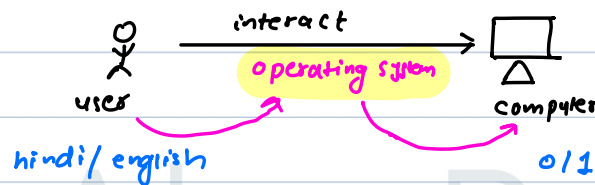


Operating system

How the applications runs behind the scene.

what is operating system?



- it acts as an intermediary between computer hardware and user application.
- its provide interface to user to interact with machines.

eg:-

window

Android

mac OS

iOS

linux

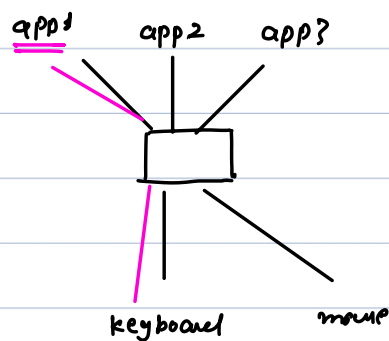
ubuntu

manager

- handle all employee and monitor them.
- ensure the equipment or material is available when need.
- allocates task to employee.

similarly, operating system manages computer resources. such as CPU, memory.

- manages workflow
- it ensure that task is completed or deadline is met.
- it manages processes.

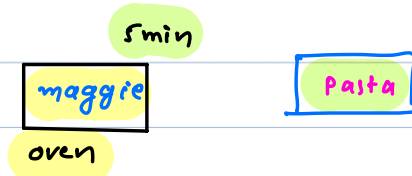


resolve conflicts among applications.

uni programming

↓
single program

it can execute only a single program at a time.

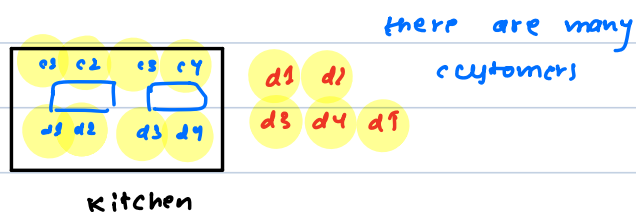


- once a program is starts executing. it continues until completion or terminated.
- easy to implement
- limiting productivity and efficiency.

multi programming:

↓
more than one

it allows to multiple programs to run concurrently on a single computers.



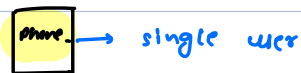
- Resource utilization
- it takes less time.
- increases efficiency and performance

sub categories of multiple prog.

single user

it is designed to supports only single user at a time.

you have the freedom to customise setting, installed applications.



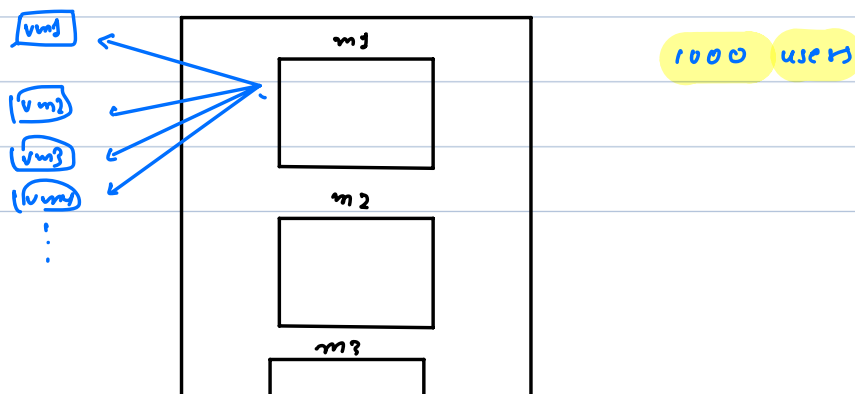
multiple user

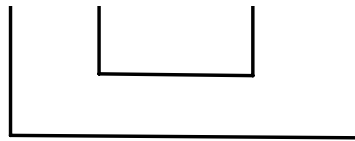
it is designed to supports multiple users simultaneously.

eg: social network Platform

window server → it allows multiple users to share resources and services.

AWS data centre





Arya

harsh

lokesh

ananya

↓
vm1

↓
vm2

↓
vm3

↓
vm4

data remains separate and isolated with others

Single
user

multiple
user

OS

OS

user1

user1

user2

user3

user4

app1

app2

app3

app4

app1

app2

app3

app4

Preemptive operating system

How OS will run multiple program

run one after one

P1

P2

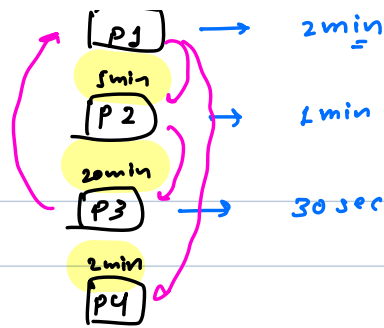
P3

P4

first P1 is execute and all are waiting.
after completion of P1, P2 will execute and
so on.

run program concurrently

10 min



(Partial part will be execute)

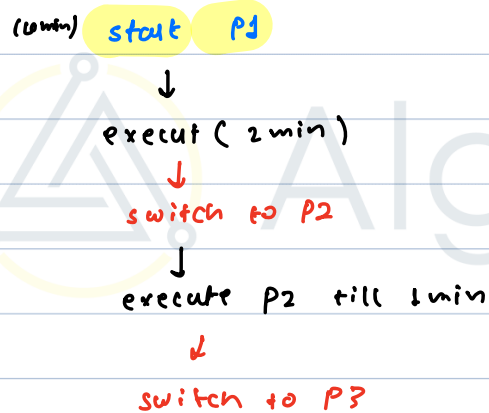
switching between programs

preemptive

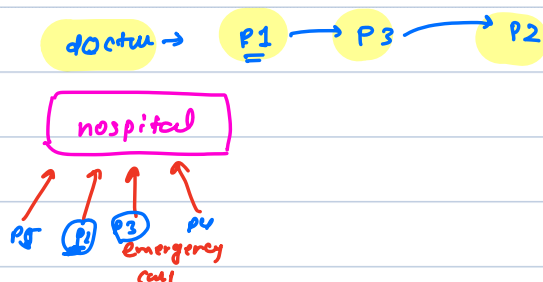
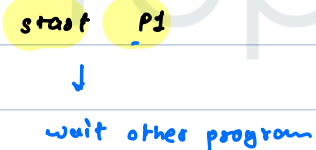
pause/ switching a program even if that not completed

non preemptive

• first we execute the program completely then we move to other program.



• we have to wait until the prev program is not completed.

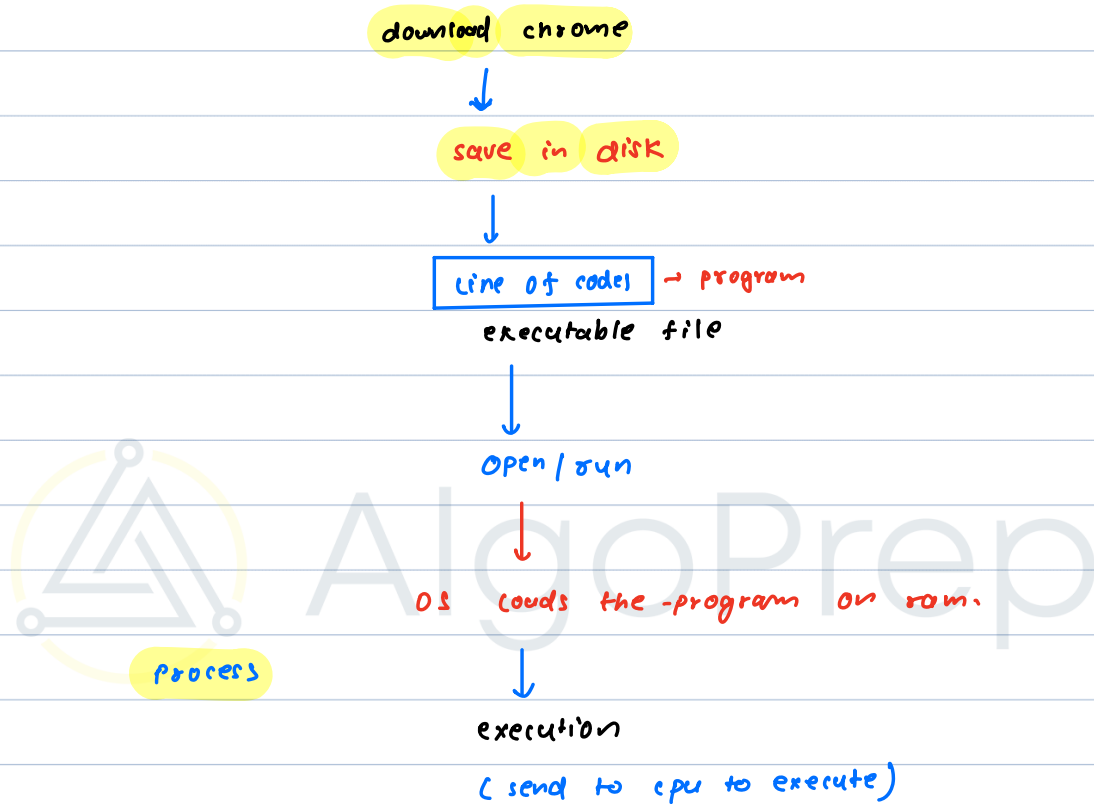


break

till

9:18

Process



- a process refers to an instance of program that is being executed by CPU.

In a computer we have multiple process

Process control block (PCB)

Pid → it is used to uniquely identify a process.

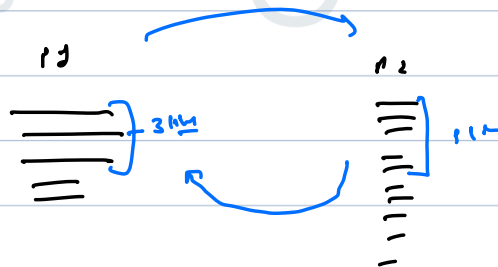
process state → running, waiting, ready

Program counter → next executable instruction.

memory manage

CPU usage

priority



```
class process {  
    int pid  
    int priority  
    int state  
    :  
}
```

Process can be multiple types

1) I/O bound process:

these processes heavily rely on I/O operations

- waiting for I/O operation till completed.
CPU is idle

eg: printer, file transfer.

2) CPU bound process:

this is mainly involves computation and algorithms operation that heavily utilizes CPU.

scientific simulation, numerical calculations.

- minimal I/O overhead.

3) mixture (CPU-bound and I/O bound) process.

- these type of process combine CPU bound task and I/O task, its involving both computational calculation and I/O operations.
- optimis resource utilization.

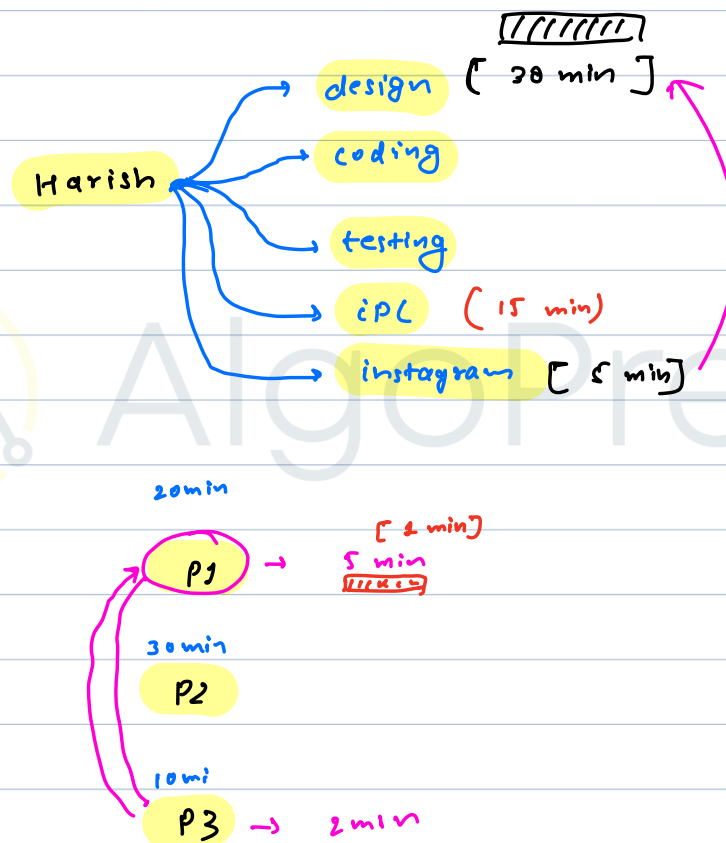
CPU scheduling → preemptive

P1

P2

P3

P4



it takes some time to recollect the information.

Context switching

if you are switching a lot between program.

CPU overhead

it loads into the memory.
execute the program till t time,
saves the information.
switch to next program



AlgoPrep